A First Course in Abstract Algebra

Section 55: Cyclotomic Extensions

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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55 Cyclotomic Extensions

Exercise 55.1: Multiplying the primitive eighth-root factors

Let ζ have order 8, so $\zeta^4 = -1$ and $\zeta^8 = 1$. The primitive roots are $\zeta, \zeta^3, \zeta^5 = -\zeta, \zeta^7 = -\zeta^3$. Pairing opposite roots gives

$$\begin{aligned}\Phi_8(x) &= (x^2 - \zeta^2)(x^2 - \zeta^6) \\ &= x^4 - (\zeta^2 + \zeta^6)x^2 + \zeta^8 = x^4 + 1,\end{aligned}$$

because $\zeta^6 = -\zeta^2$. This computes all coefficients rather than just identifying the zeros.

Exercise 55.2: The twentieth-root group as an abelian group

The group is $(\mathbb{Z}/20\mathbb{Z})^\times$. Since $20 = 4 \cdot 5$ with coprime factors, the Chinese remainder theorem gives

$$(\mathbb{Z}/20\mathbb{Z})^\times \cong (\mathbb{Z}/4\mathbb{Z})^\times \times (\mathbb{Z}/5\mathbb{Z})^\times \cong \mathbb{Z}_2 \times \mathbb{Z}_4.$$

The factors have orders 2 and 4, so the group has order 8. This is its invariant-factor classification because $2 \mid 4$.

Exercise 55.3: Three Euler totients

For $n = \prod p_j^{e_j}$, $\varphi(n) = \prod p_j^{e_j - 1}(p_j - 1)$, since a residue must not be divisible by any prime divisor of n.

(a) $60 = 2^2 \cdot 3 \cdot 5$, so $\varphi(60) = 2 \cdot 2 \cdot 4 = 16$.

(b) $1000 = 2^3 \cdot 5^3$, so $\varphi(1000) = 4 \cdot 100 = 400$.

(c) $8100 = 2^2 \cdot 3^4 \cdot 5^2$, so $\varphi(8100) = 2 \cdot 54 \cdot 20 = 2160$.

Exercise 55.4: The first thirty constructible regular polygons

The criterion is $n = 2^a$ times a product of distinct Fermat primes. Enumerating these numbers increasingly, starting at $n \geq 3$, gives

$$\begin{gathered}3, 4, 5, 6, 8, 10, 12, 15, 16, 17, \\ 20, 24, 30, 32, 34, 40, 48, 51, 60, 64, \\ 68, 80, 85, 96, 102, 120, 128, 136, 160, 170.\end{gathered}$$

For completeness of the enumeration up to 170, its only possible odd prime factors are 3,5,17; the next Fermat prime is 257. No square of an odd prime is allowed. Listing $2^a d \leq 170$ for $d \in \{1, 3, 5, 15, 17, 51, 85\}$, omitting 1 and 2 and sorting, gives exactly the displayed thirty numbers.

Exercise 55.5: The smallest constructible integral-degree angle

The answer is $3°$. A regular 120-gon is constructible because $120 = 2^3 \cdot 3 \cdot 5$, so its central angle $360°/120 = 3°$ is constructible. A $1°$ angle would construct a regular 360-gon, but $360 = 2^3 \cdot 3^2 \cdot 5$ contains a square of the odd prime 3. A $2°$ angle would similarly construct a regular 180-gon, and $180 = 2^2 \cdot 3^2 \cdot 5$ fails the same criterion. Thus neither smaller positive integral angle is constructible.

Exercise 55.6: Twelfth roots and a group of exponent two

(a) The splitting field is $K = \mathbb{Q}(\zeta_{12})$, of degree $\varphi(12) = 4$ over $\mathbb{Q}$.

(b) Its automorphisms are the exponent maps for $1, 5, 7, 11$ modulo 12. Each exponent squares to 1 modulo 12, so every automorphism squares to the identity. A group of order 4 with no element of order 4 is $\mathbb{Z}_2 \times \mathbb{Z}_2$, giving the requested classification.

Exercise 55.7: Two cyclotomic polynomials in finite characteristic

Over $\mathbb{Z}_2$, the primitive third roots are the nonidentity roots of $x^3 - 1$, so $\Phi_3(x) = x^2 + x + 1$. Over $\mathbb{Z}_3$, the same opposite-root computation as Exercise 55.1 gives $\Phi_8(x) = x^4 + 1$. It factors there as $(x^2 + x + 2)(x^2 - x + 2)$; each quadratic has no root in $\mathbb{Z}_3$. This illustrates that a cyclotomic polynomial need not be irreducible over a finite field.

Exercise 55.8: A repeated polynomial already split over the prime field

In characteristic 3,

$$x^6 - 1 = (x^2 - 1)^3 = (x - 1)^3(x + 1)^3.$$

Its only distinct roots are $1, -1$, already in $\mathbb{Z}_3$. Therefore the splitting field is $\mathbb{Z}_3$ itself, with 3 elements. Counting the formal degree as the number of distinct roots would give the wrong answer.

Exercise 55.9: Ten statements about cyclotomic fields

(a) False. Over $\mathbb{C}$, for instance, every cyclotomic polynomial of degree greater than 1 splits into linear factors. Irreducibility here is over $\mathbb{Q}$.

(b) True. These are exactly its defining zeros.

(c) False. The group has order $\varphi(n)$; for example $n = 3$ gives order 2, not 3.

(d) True. It is isomorphic to the multiplicative group of units modulo n.

(e) True. $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$ and the extension is Galois.

(f) False. $25 = 5^2$ has a squared odd prime factor; also $\varphi(25) = 20$ is not a power of 2.

(g) True. $17 = 2^{2^2} + 1$ is a Fermat prime.

(h) True for an ordinary polygon, so prime $p \geq 3$: the criterion says an odd prime p must be Fermat. If one includes the degenerate $p = 2$ case, it must be treated separately; 2 is not a Fermat prime.

(i) False. $2^{2^5}+1 = 4294967297 = 641 \cdot 6700417$ is composite.

(j) True by definition: a Fermat prime is a prime among the numbers $2^{2^k}+1$.

Exercise 55.10: Partitioning roots by their orders

Assume the characteristic does not divide n. The derivative of $x^n - 1$ is nx^{n-1}, so all its roots in a closure are distinct. Each root has a unique multiplicative order d dividing n, and is then a primitive dth root. The root set is therefore the disjoint union of the primitive-root sets for all $d \mid n$. Multiplying their monic linear-factor products gives

$$x^n - 1 = \prod_{d|n} \Phi_d(x).$$

All the cyclotomic factors have coefficients in the prime field, hence in F, so this is an identity in $F[x]$.

Exercise 55.11: The first six cyclotomic polynomials

Use $x^n - 1 = \prod_{d|n} \Phi_d(x)$ and divide by factors from proper divisors. This gives

n	$\Phi_n(x)$
1	$x-1$
2	$x+1$
3	x^2+x+1
4	x^2+1
5	$x^4+x^3+x^2+x+1$
6	x^2-x+1

For example, $x^6 - 1 = (x-1)(x+1)(x^2+x+1)(x^2-x+1)$, so removing the factors for divisors 1,2,3 leaves the factor for 6.

Exercise 55.12: The twelfth cyclotomic polynomial

The divisors of 12 are $1, 2, 3, 4, 6, 12$. An efficient cancellation is

$$x^{12} - 1 = (x^6 - 1)(x^6 + 1), \qquad x^6 + 1 = (x^2 + 1)(x^4 - x^2 + 1).$$

The first factor already contains $\Phi_1, \Phi_2, \Phi_3, \Phi_6$, and $x^2 + 1 = \Phi_4$. Thus the remaining factor is $\Phi_{12}(x) = x^4 - x^2 + 1$.

Exercise 55.13: Negating roots of odd order

If $n > 1$ is odd and ζ has order n, then $-\zeta$ has order $2n$: its nth power is -1, and if $(-\zeta)^k = 1$, then squaring gives $n \mid k$ and the odd multiple $k = n$ does not work. Conversely, negating a primitive $2n$th root gives a primitive nth root. Thus the zeros of $\Phi_n(-x)$ are exactly those of $\Phi_{2n}(x)$. For odd $n > 1$, $\varphi(n)$ is even, since invertible residues pair as $a, -a$, so $\Phi_n(-x)$ is monic. The two monic polynomials with the same distinct roots coincide.

Exercise 55.14: The compositum for coprime root orders

Let ζ have order nm, and set $\xi = \zeta^m$, $\eta = \zeta^n$, which have orders n, m. Their fields are contained in $\mathbb{Q}(\zeta)$. Conversely choose integers u, v with $um + vn = 1$. Then $\zeta = \xi^u \eta^v \in \mathbb{Q}(\xi, \eta)$. Thus

$$\mathbb{Q}(\zeta_{nm}) = \mathbb{Q}(\zeta_n, \zeta_m),$$

which is exactly the splitting field of $(x^n - 1)(x^m - 1)$.

Exercise 55.15: The product decomposition for coprime orders

Identify the three cyclotomic Galois groups with unit groups modulo nm, n, m. Restriction to the two smaller splitting fields is the map

$$(\mathbb{Z}/nm\mathbb{Z})^\times \longrightarrow (\mathbb{Z}/n\mathbb{Z})^\times \times (\mathbb{Z}/m\mathbb{Z})^\times, \quad a \longmapsto (a \bmod n, a \bmod m).$$

It is injective because equality modulo both coprime moduli implies equality modulo their product. The Chinese remainder theorem gives a unique residue modulo nm for each pair of unit residues, and that residue is also a unit, proving surjectivity. Multiplication is preserved. Therefore the group of $x^{nm} - 1$ is the direct product of the two smaller groups, as claimed.